[ARCHIVED CATALOGUE]

# Applied Physics (MS)

The MS degree in Applied Physics is directed toward students with interests in engineering physical systems for which acquisition and interpretation of information requires an understanding of underlying physical models, measurement techniques and disruptive factors such as noise, turbulence, and unwanted signals. Examples include RF electromagnetic or acoustic detection of objects or resources, photonic sources and detection and applications that involve the mechanics of solid and fluid media.

## Required Analytical Courses

- AME 508 Machine Learning and Computational Physics Units: 4
- EE 604 Computational Methods in Applied Physics Units: 4

## Topical Areas

Take at least **two core courses** (8 units) from the topical areas below, and take **three elective courses** (12 units) that follow the core courses chosen. Other courses may be substituted for listed electives subject to the approval of the program adviser.

A thesis (AME 594a, AME 594b or EE 594a, EE 594b) can substitute for 4 elective units.

Students may also take up to two units of directed research (AME 590 or EE 590) to fulfill the 28-unit degree requirement.

### Electromagnetic Wave Propagation and Scattering

- EE 570a Advanced Electromagnetic Theory Units: 4

Elective courses:

- EE 551 Principles of Radar Units: 3
- EE 570b Advanced Electromagnetic Theory Units: 4
- EE 571 Wave Interactions with Random and Inhomogeneous Media Units: 4
- EE 573a Antenna Systems Engineering Units: 4
- EE 573b Antenna Systems Engineering Units: 4
- EE 578 Computational Electromagnetics for Engineers Units: 4

### Optics and Photonics

- EE 530 Optical Materials, Instruments and Devices Units: 4

Elective courses:

- EE 529 Optics Units: 4
- EE 531 Nonlinear Optics Units: 4
- EE 539 Engineering Quantum Mechanics Units: 4
- EE 540 Introduction to Quantum Electronics Units: 4
- EE 566 Optical Information Processing Units: 4

### Mechanics of Fluid and Solid Media

- AME 506 Continuum Mechanics Units: 4 or
- AME 509 Applied Elasticity Units: 4

Electives courses:

- AME 511 Compressible Gas Dynamics Units: 4
- AME 513a Fundamentals and Applications of Combustion Units: 4
- AME 513b Fundamentals and Applications of Combustion Units: 4
- AME 515 Thermal and Biological Transport Phenomena I Units: 4
- AME 521 Engineering Vibrations II Units: 4
- AME 530a Dynamics of Incompressible Fluids Units: 4
- AME 530b Dynamics of Incompressible Fluids Units: 4
- AME 630 Transition to Chaos in Dynamical Systems Units: 4